Acme Aerotech

Network Security & Resilience Assessment

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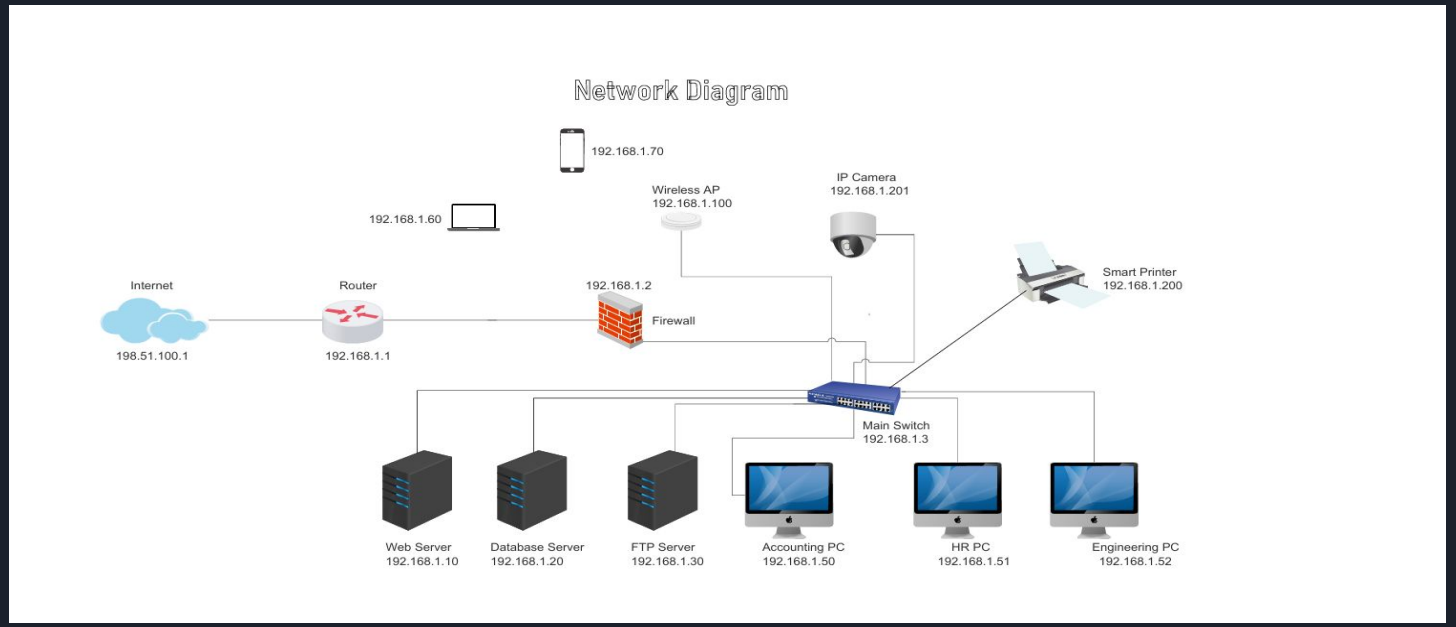
Organizational Context & Scope

- Aerospace parts manufacturer
- ~ 75 employees
- Small IT team
- Government contract requires stronger cybersecurity
- First formal network assessment

Original Network Overview



- Flat network topology
- Minimal segmentation
- Single firewall, router and switch
- Users, servers and IoT devices on the same LAN

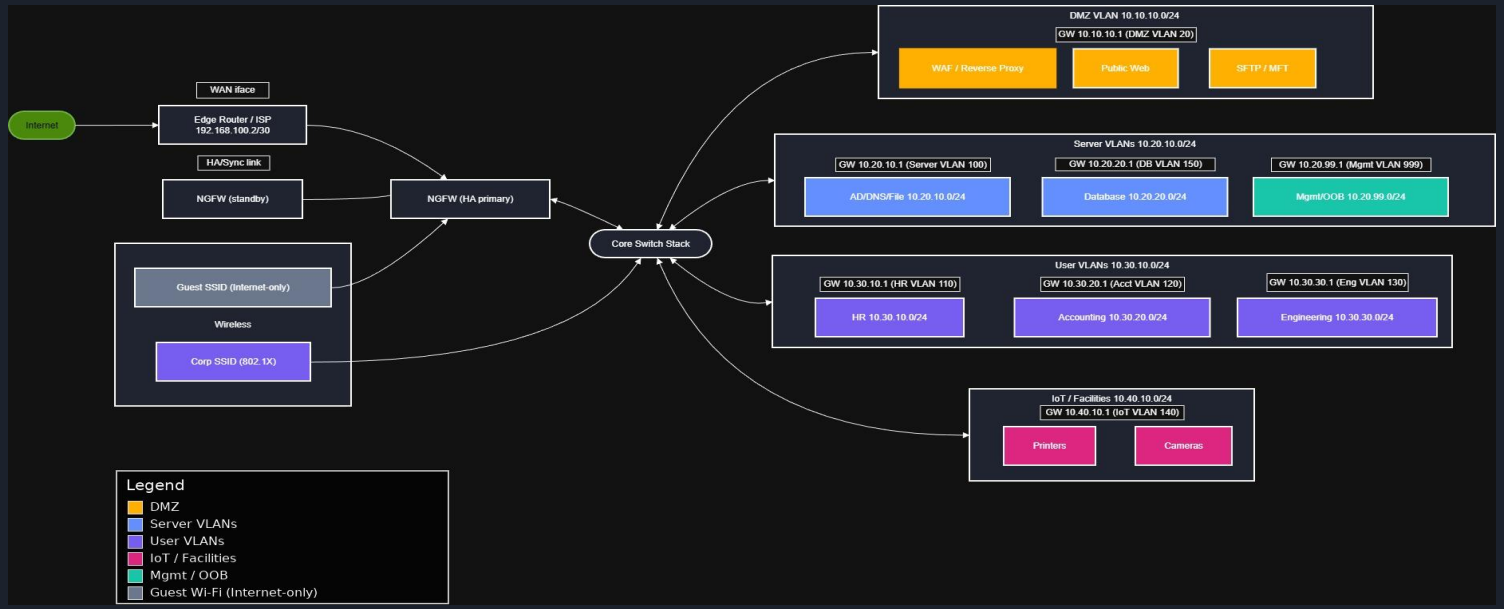
Vulnerabilities & Availability Risks

- Flat network = lateral movement
- Default credentials on critical systems
- No patch management
- Public services inside LAN
- Single points of failure
- IoT devices on production network

Design Goals

- Enforce least privilege
- Separate trust zones
- Protect public services
- Reduce single points of failure
- Improve maintainability

Redesigned Network Architecture



- Segmented VLAN architecture
- Dedicated DMZ
- Firewall high availability
- Core switch redundancy
- Clear trust boundaries

Security & Resilience Improvements

- VLAN separation by role
- DMZ with reverse proxy
- Secure wireless (802.1X + guest isolation)
- IoT and management isolation
- Reduced blast radius

Reflection: What I Learned

- Importance of segmentation
- Security and availability are linked
- Value of clear diagrams

Challenges & What I'd Do Differently

- Balancing security vs. simplicity
- VLAN planning and labeling
- Next time: cost analysis, monitoring, cloud options

Conclusion

- Risks identified
- Secure and resilient redesign
- Business requirements met